

Name: _____

Student Number: _____

ELECENG 3TP3 C01
Signals and Systems

DAY CLASS

Prof. T. Todd

DURATION OF EXAMINATION: 2.5 Hours

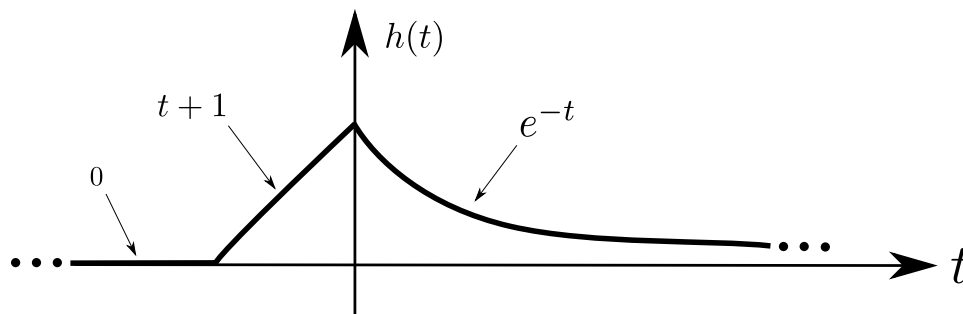
MCMASTER UNIVERSITY FINAL EXAMINATION

DECEMBER 2021

THIS EXAMINATION PAPER INCLUDES 6 PAGES AND 6 QUESTIONS. YOU ARE RESPONSIBLE FOR ENSURING THAT YOUR COPY OF THE PAPER IS COMPLETE. BRING ANY DISCREPANCY TO THE ATTENTION OF YOUR INVIGILATOR.

Special Instructions: Open book examination.
Answer all questions either on the PDF file provided or on your own pages.
Upload your answers to your Final Exam drop box on Avenue To Learn.
Those without SAS accommodations must upload your answers by 11:45am
In all questions be sure to explain any assumptions you make.

- (20) 1. Consider the function $h(t)$ shown below.



- (a) For what values of t_0 would the function $h(t + t_0 + 1)$ be a possible impulse response of a causal linear time invariant (LTI) system?

(5) when $t_0 + 1 \leq -1$ or $t_0 \leq -2$

- (b) For what range of values of t is the convolution, $h(t) * h(t)$, equal to zero?

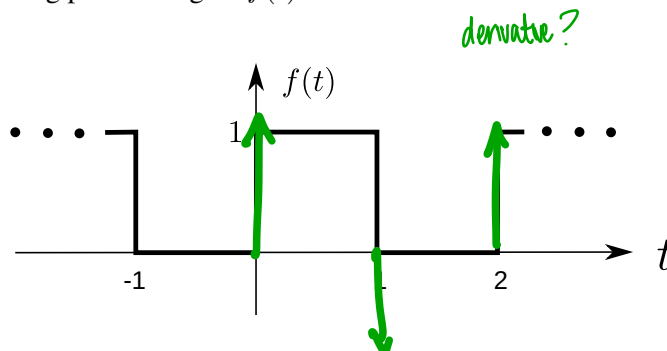
(5) $t \leq -2$

- (c) Write out the expression (i.e., integral(s)) that you would need to solve in order to obtain the convolution of $h(t)$ with itself (i.e., $h(t) * h(t)$) for $t \geq 0$. Do not solve this expression, just write out any integrals, etc.

(10) $\int_{-1}^0 (\tau + 1)e^{\tau-t} d\tau + \int_0^t e^{-\tau} e^{\tau-t} d\tau + \int_t^{t+1} (1 + t - \tau)e^{-\tau} d\tau$

Part marks if the procedure is right but some details are wrong, e.g., labeling the front of the shifted function wrong, but the rest seems correct.

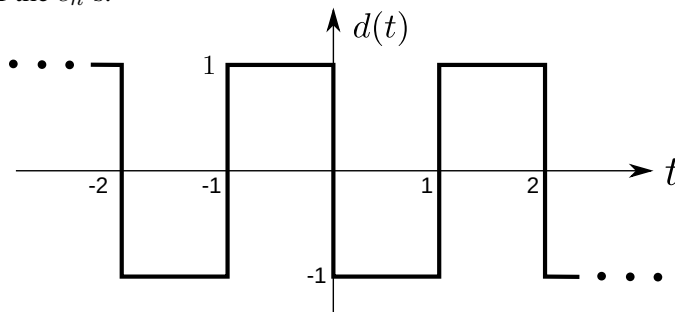
- (20) 2. Consider the following periodic signal $f(t)$ where the time axis values are in seconds.



- (a) Show how you can use the derivative property of the Fourier series to obtain all the complex exponential Fourier series coefficients, c_n , for $f(t)$, i.e., $f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$. Write the final expression in terms of the function $\text{sinc}(x) = \sin(\pi x)/\pi x$.

(5) Note that $\omega_0 = \pi$. Taking the derivative of $f(t)$ over one period gives $f'(t) = \delta(t) - \delta(t-1)$. Therefore $d_n = \frac{1}{2} \int_T (\delta(t) - \delta(t-1)) e^{-jn\pi t} dt$. Evaluating this integral and dividing by $jn\pi$ gives $c_n = \frac{e^{-jn\pi/2}}{2} \text{sinc}(n/2)$ for $n \neq 0$. $c_0 = 1/2$.

- (b) Given that you know the c_n 's of $f(t)$ from Part 2a, use them to find the Fourier series coefficients, d_n , of $d(t)$ given below. Do not substitute your answer from Part 2a into your result, just leave it in terms of the c_n 's.

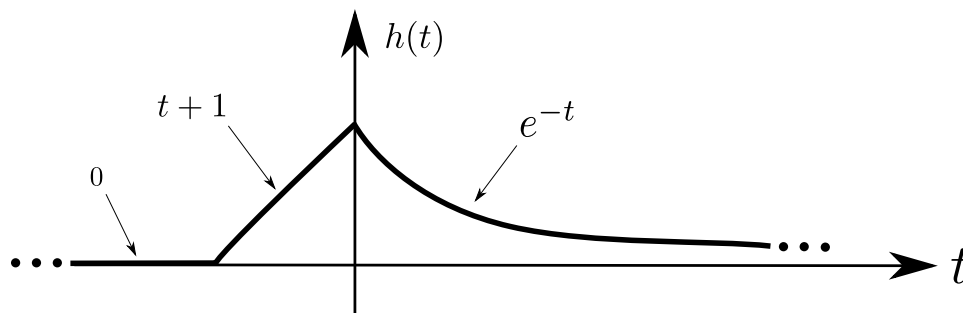


(5) Note that $d(t) = 2f(t-1) - 1$. Therefore, $d_n = 2e^{-jn\omega_0} c_n = 2e^{-jn\pi} c_n = 2(-1)^n c_n$ for $n \neq 0$ and $d_0 = 0$.

- (c) Assume that $f(t)$ from Part 2a is input to a linear time-invariant system whose impulse response is given by $h(t) = e^{-t}u(t-a)$ where a is a positive real number. How is the phase of the fundamental frequency component of $f(t)$ affected after $f(t)$ passes through the system?

(10 = 5 for FT + 5 for phase.) $e^{-t}u(t-a) = e^{-a} \cdot e^{-(t-a)}u(t-a) \longleftrightarrow \frac{e^{-a} \cdot e^{-j\omega}}{1+j\omega} = \frac{e^{-a(1+j\omega)}}{1+j\omega}$.
The fundamental frequency is $\omega_0 = \pi$. Therefore $H(\pi) = \frac{e^{-a(1+j\pi)}}{1+j\pi}$. The phase is changed by $-a\pi - \tan^{-1}(\pi)$

- (15) 3. Consider the signal $h(t)$ from Question 1 shown below.



Find the Fourier transform, $H(\omega)$, of $h(t)$. To help obtain the answer, use the fact that the Fourier transform of a rectangular pulse

$$p_{\tau}(t) = \begin{cases} 1 & |t| \leq \tau/2 \\ 0 & \text{otherwise} \end{cases}$$

is given by

$$P_{\tau}(\omega) = \tau \operatorname{sinc}\left(\frac{\tau\omega}{2\pi}\right)$$

Be sure to state any Fourier transform properties that you use.

One way is to write $h(t) = x(t) + e^{-t}u(t)$ where $x(t)$ is the triangle pulse to the left of the origin. Note that $e^{-t}u(t) \longleftrightarrow \frac{1}{1+j\omega}$. Taking the derivative, $x'(t) = p_1(t + 1/2) - \delta(t)$ we have that $x'(t) \longleftrightarrow \operatorname{sinc}(\omega/2\pi)e^{j\omega/2} - 1$. Using the derivative property, $X(\omega) = \frac{1}{j\omega} \operatorname{sinc}(\omega/2\pi)e^{j\omega/2} - 1$. Then $H(\omega) = \frac{1}{j\omega} \operatorname{sinc}(\omega/2\pi)e^{j\omega/2} - 1 + \frac{1}{j\omega+1}$. Note that there are other minor variations.

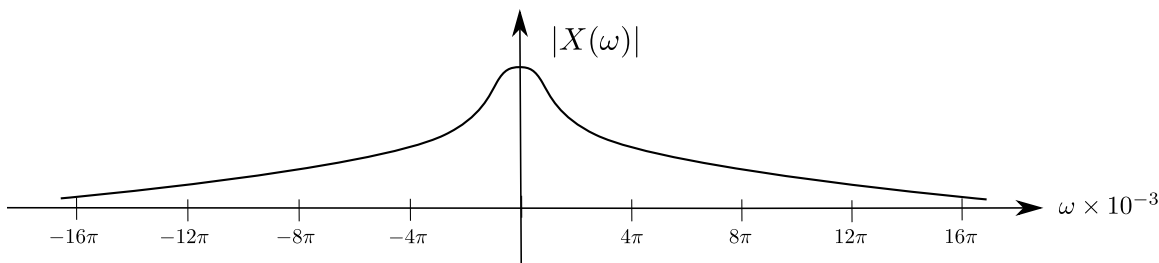
- (15) 4. In Laboratory 3 you sampled some sinusoids and observed the reconstructed waveforms. Answer the following questions.

- (a) A sinusoid of frequency 13 KHz is periodically sampled. When the samples are reconstructed, it appears as a 3 KHz sinusoid. Explain, in detail, what has happened and determine what sampling frequency, f_s in Hz, was used. Describe the details of any theory that you have used to arrive at your answer.

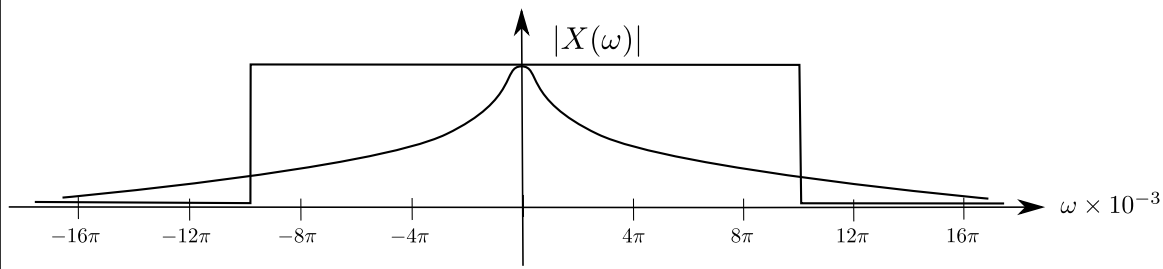
(10) The sampling frequency was below the Nyquist rate (i.e., $2 \times f_{\max}$). Therefore, the sinusoid has been aliased to 3 KHz and the sampling rate is $f_s = 16$ KHz. The sinusoid is translated up and down by multiples of 16 KHz, giving the 3 KHz result. They should mention the Sampling Theorem.

- (b) The following graph is the magnitude of the Fourier transform, $X(\omega)$, of a signal $x(t)$ that is to be sampled at a rate of 10 KHz. The ω axis is in units of $\text{rad/sec} \times 10^{-3}$, e.g., the 8π point is 8000π rad/sec.

Determine the cutoff frequency of the anti-aliasing filter needed (in rad/sec), i.e., the frequency beyond which only a negligible amount of signal would pass so that aliasing is largely prevented. On the diagram below, sketch what the filter response magnitude would be assuming an ideal low pass filter response for your sketch. Explain in detail how you arrived at your answer.

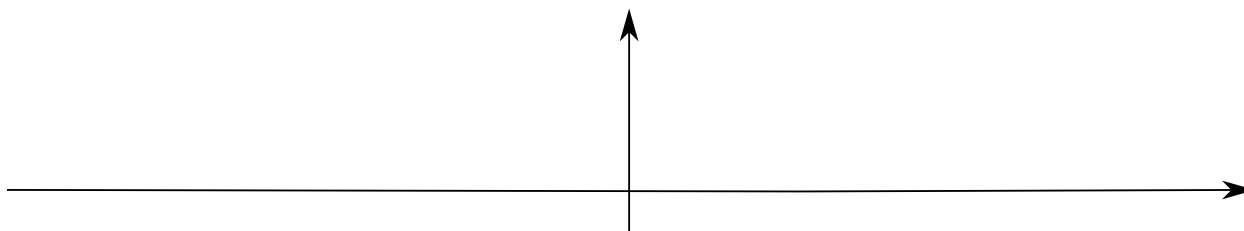


(5) The filter should cut off by at least 5 KHz to avoid aliasing, i.e., $\omega_c = 2\pi \times 5000 = 10\pi \times 10^3$. 10π on the graph. Some students show the filtered version of $X(\omega)$ which is ok.



(10) 5. Answer the following question relating to the DTFT and the DFT.

- (a) A signal, $s(t) = \cos(\omega_1 t) + \cos(\omega_2 t)$, is periodically sampled at a frequency of 32 KHz, where $\omega_1 = 8\pi \times 10^3$ and $\omega_2 = 16\pi \times 10^3$ rad/sec. This creates a discrete-time signal $s[n] = s(nT)$ where T is the sampling period. On the graph below, sketch the magnitude of the discrete-time Fourier transform (DTFT), $S(\Omega)$, of $s[n]$, where $\Omega = \omega T$. Make sure to label the values on both axes.

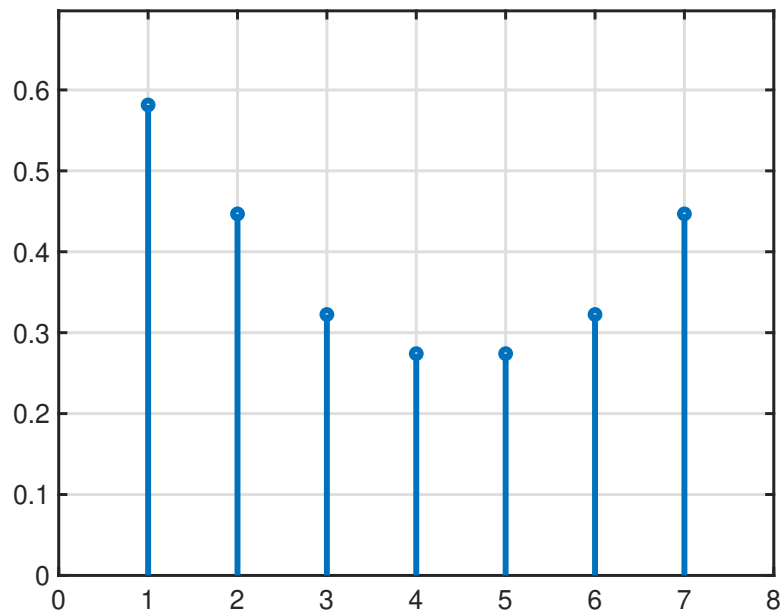


(5) Using the given parameters, the sinusoid frequencies are $\Omega_1 = \omega T = \pi/4$ and $\Omega_2 = \omega T = \pi/2$. From the DTFT table, there are therefore impulse pairs at $\pm\pi/4$ and $\pm\pi/2$, each of magnitude π . These impulses are repeated at all negative and positive multiples of π .

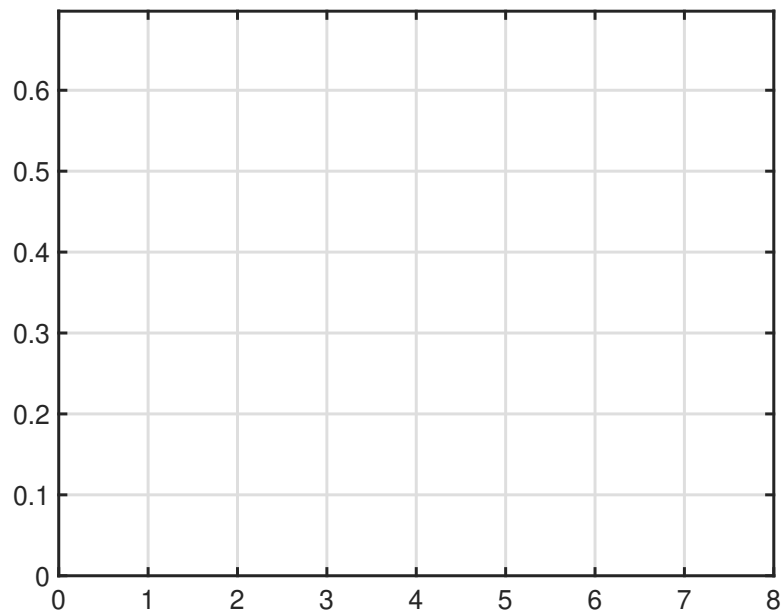
- (b) A discrete-time signal, $x[n]$, is stored in a 1×7 MATLAB vector x . The FFT is taken and it's magnitude is plotted using the MATLAB statements, $X = \text{fft}(x)$ and $\text{stem}(\text{abs}(X))$. Some other syntax is used for formatting the plot, which is shown below.

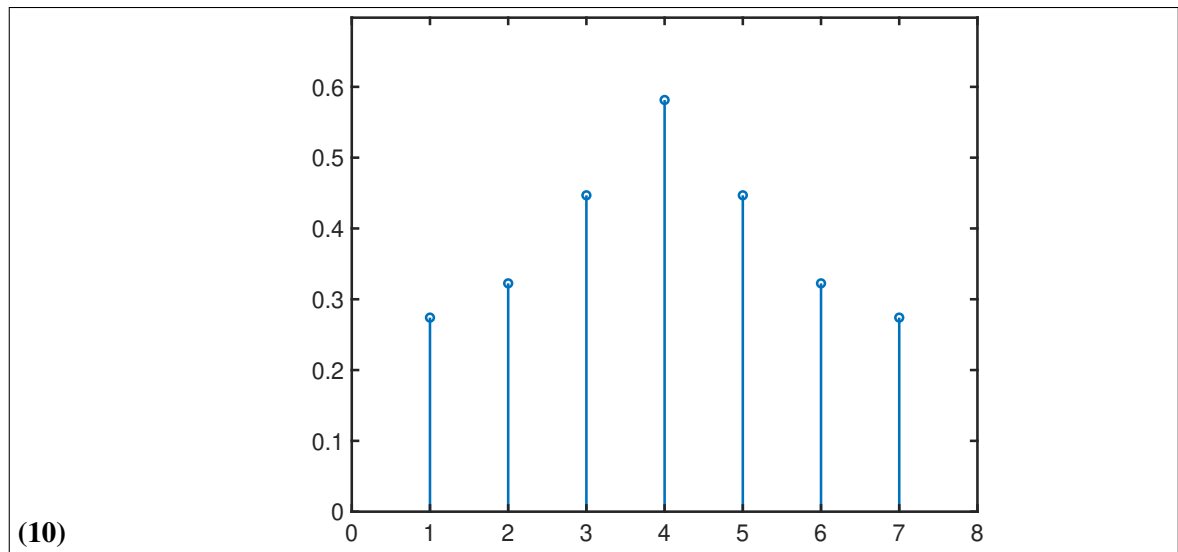
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Sketch what this graph would be if the second MATLAB statement is changed to `stem(abs(fftshift(X)))`, i.e., show what effect the `fftshift` function has.





- (20) 6. Consider a discrete-time LTI system with input $x[n]$ and output $y[n]$ defined by

$$y[n+1] - y[n] = x[n+1] + x[n]$$

Find the Z transform of the complete response (i.e., zero-state plus zero-input responses) of the system when the input is given by

$$x[n] = \left(-\frac{1}{2}\right)^{n-2}$$

for $n \geq 0$ and where the initial conditions are $x[-1] = y[-1] = \frac{1}{2}$.

10 marks for rewriting the difference equation as $y[n+1] - y[n] = x[n+1] + x[n]$ and taking its Z transform that includes the initial conditions, i.e., $Y[z] = X(z)\frac{z+1}{z-1} + (x[-1] + y[-1])\frac{z}{z-1}$ or $Y[z] = X(z)\frac{z+1}{z-1} + \frac{z}{z-1}$

10 marks for getting the Z transform of $x[n]$ and incorporating it into their first result, i.e., $X(z) = \frac{4z}{z+\frac{1}{2}}$. Therefore $Y[z] = \frac{4z}{(z+1)}\frac{z+1}{z-1} + \frac{z}{z-1}$

THE END